From OpenAI to Open Source AI

Navigating Between Commercial Ownership and Collaborative Openness



Raphaël Semeteys (and Luxin Zhang) - Worldline

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Introduction



Raphaël Semeteys

- Open source since 1997, professionally since 2004
- Yoga Teacher, Creator of the QSOS method
- Head of DevRel at **Worldline**

We design **payments technology** that powers the growth of millions of businesses around the world

- 7000+ engineers in over 40 countries
- Managing 43+ billion transactions per year
- €250M spent in R&D every year
- Handling 150+ payment methods

The **early days** of LLMs

From rule-based and simpler statistical models to LLMs



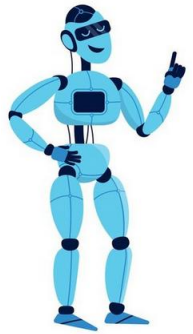
2010's

Word embeddings such as Word2Vec and GloVe



2017-2018

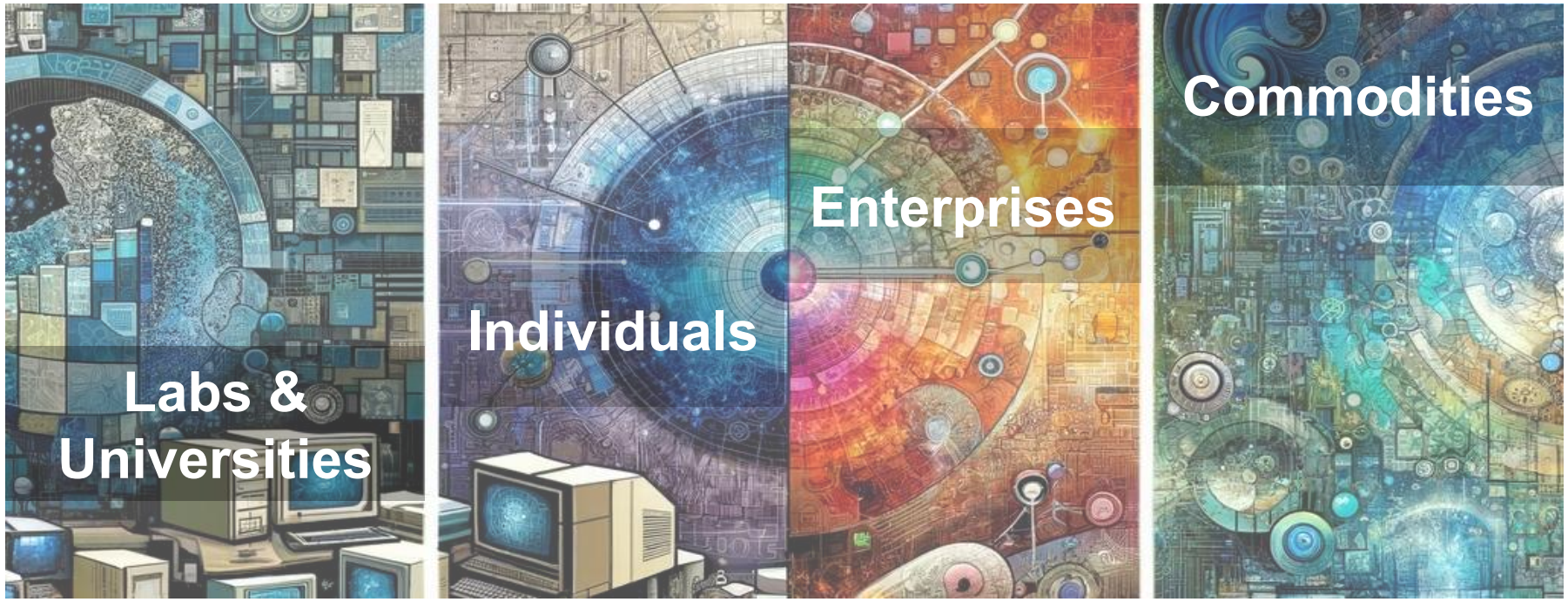
"*Attention is All You Need*" Transformers, BERT



2020's

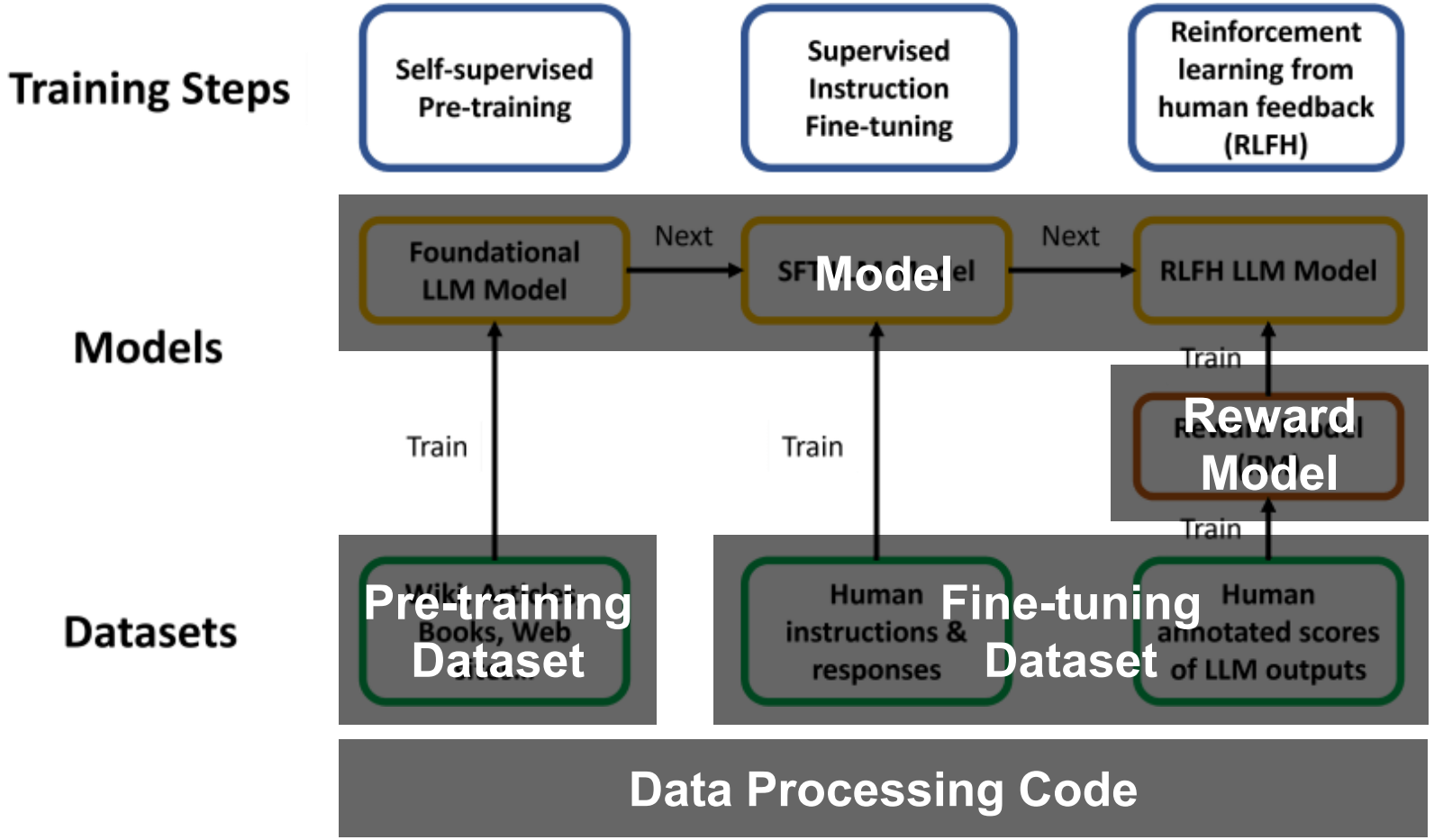
Generative AI, ChatGPT responsibility concerns

GenAI is having its **Linux Moment**



- Just like open source and Internet, bust much faster!
- Dynamics between collaborative openness and commercial ownership
- Need of clarity on licenses

Defining **Openness** of a LLM



Defining **Openness** of a LLM

Score	Level	Description	Model (weights)	Pre-training Dataset	Fine-tuning Dataset	Reward model	Data Processing Code
0	**Closed**	No access to any public information, data or asset					
1	**Published research only**	Research papers(s) published but with no more information, data or asset					
2	**Restricted access**	Access to asset is possible only with special agreement (commercial, research…)					
3	**Open with limitations**	Access and reuse of asset is possible but with certain limitations on usage (ex. Open RAIL)					
4	**Totally open**	Access and reuse of asset is possible without restriction (ex. open source license)					

Market-Leading Player: OpenAI

Deviation from original vision of research transparency & openness

GPT-1 & 2

Non/For-profit (US)		
Component	**Score**	**Level description**
Model	4	Totally open
Dataset	1	Published research only
Code	1	Published research only

→

GPT-3 & 4

0	Closed

ChatGPT

research paper only

No training of other commercial LLMs

Market-Leading Player: OpenAI

Deviation from original vision of research transparency & openness

GPT-1 & 2

GPT-3 & 4

OpenAI GPT

Non/For-profit (US)		
Component	**Score**	**Level description**
Model	4	To talk about
Code	1	Published research only

0	Closed

You may not: [...] Use Output to develop models that compete with OpenAI.

ChatGPT

research paper only

No training of other commercial LLMs

Market-Leading Player: Google

Transition from open research to proprietary commercial approach

BERT

Enterprise (US)		
Component	**Score**	**Level description**
Model	4	Totally open
Dataset	2	Restricted access
Code	4	Totally open

→

PaLM 2 & Gemini

1	Published research only
1	Published research only
0	Closed

Market-Leading Player: Meta

Journey to openness

RoBERTa

Enterprise (US)		
Component	**Score**	**Level description**
Model	4	Totally open
Dataset	3	Open with limitations
Code	4	Totally open

→

Llama 2

3	Open with limitations
1	Published research only
1	Published research only

Restriction on usage: license for platforms with 700+ M users

Market-Leading Player: Meta

Journey to openness

RoBERTa

Llama 2

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Restriction on usage: license for platforms with 700+ M users

Llama offspring's: Alpaca and Vicuna

Fine-tuned models from Llama 2 by universities





Research (US)		
Component	**Score**	**Level description**
Model	3	Open with limitations
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	2	Research use only
Code	4	Under Apache 2 license



Restrictions from both Llama 2 and OpenAI (ShareGPT)

Collaborative **foundational** LLMs

	Non-profit (US)		Research (UAE)		Research (EU)		Research (US)		Enterprise (FR)	
	EleutherAI GPT-J		Falcon		BLOOM		OpenLLaMa		Mistal/Mixtral	
Model	4	Access and reuse without restriction	3	Open with limitations	3	Open RAIL license	4	Access and reuse without restriction	4	Access and reuse without restriction
Dataset	3	Open with limitations	4	Access and reuse without restriction	3	Open with limitations	4	Access and reuse without restriction	0	No public information or access
Code	4	Completely open	1	General instructions	4	Completely open	1	Just examples	4	Completely open

Dataset fuzziness: *please refer to the specific license depending on the subset you use*

Notion of responsible usage

Collaborative **foundational** LLMs

	Non-profit (US)	Research (UAE)	Research (EU)	Research (US)	Enterprise (FR)
	EleutherAI GPT-J	Falcon	BLOOM	OpenLLaMa	Mistal/Mixtral
Model	4				ccess and use without estriction
Dataset	3				o public nformation or ccess
Code	4				ompletely pen

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Dataset fuzziness: *please refer to the specific license depending on the subset you use*

Notion of responsible usage

Collaborative **fine-tuned** LLMs

	Enterprise (US)		Enterprise (US)		Enterprise (US)		Consortium (UAE/US)	
	Dolly		BLOOMChat		Zephyr		LLM360	
Model	4	Based on GPT-J	3	Based on BLOOM	4	Based on Mistral	4	Open source
Pre-training Dataset	3	Based on GPT-J	3	Based on BLOOM	0	Based on Mistral	4	RedPajama, Falcon, StarCoder
Fine-tuning Dataset	4	Access and reuse without restriction	4	Dolly and LAION	2	Research use only (OpenAI)	2	Research use only (OpenAI)
Reward model	0	No public information available	0	No public information available	3	Paper and code examples	0	No public information available
Code	4	Open source	3	OpenRAIL	3	Example code available	4	Open source

Impact of foundational model or pre-training datasets

Collaborative **fine-tuned** LLMs

	Enterprise (US)		Enterprise (US)		Enterprise (US)		Consortium (UAE/US)	
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Pre-trai… Dataset								…oder
Fine-tur… Dataset								…only
Reward								
Code	4	Open source	3	OpenRAIL	3	…available	4	Open source

BLOOMChat Use Restrictions
l. To provide medical advice and medical results interpretation; or
m. To generate or disseminate information for the purpose to be used for administration of justice, law enforcement, immigration or asylum processes, such as predicting an individual will commit fraud/crime commitment.

Impact of foundational model or pre-training datasets

Collaboration **platform**: Hugging Face



Hugging Face

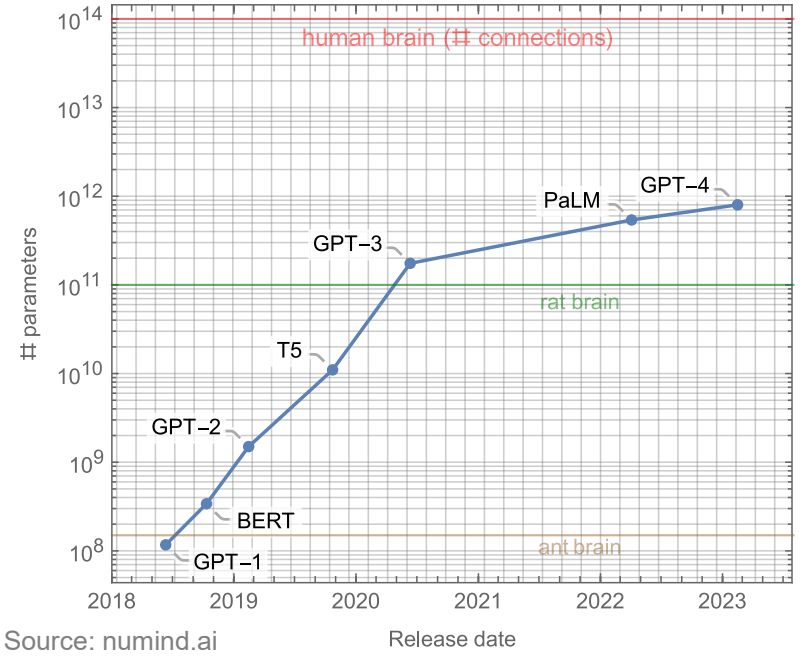
Enabler for collaboration and reuse

- Startup and ecosystem dedicated to democratizing AI
  - Open source Transformers library
  - LLM leaderboard: upload and assess models

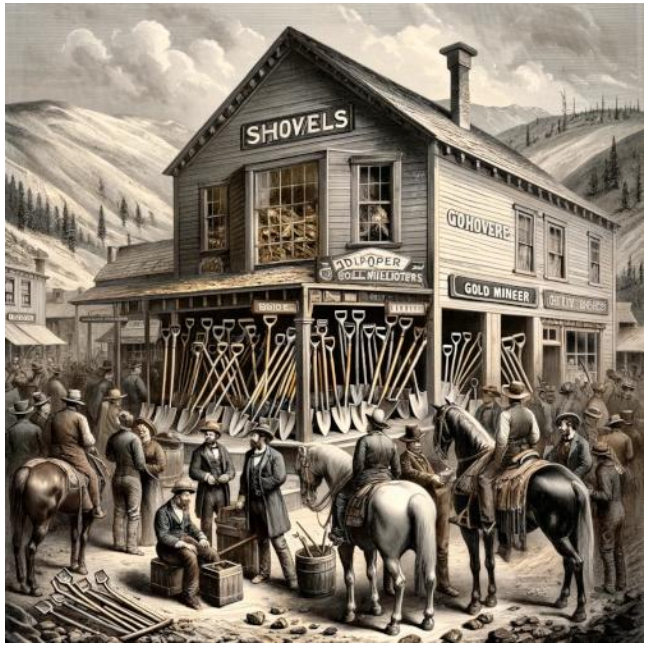
- The “GitHub of AI” Models 489,090 Datasets 102,662
  - Collaborative space for exploring, sharing and experimenting AI
  - Hosts thousands of models, datasets, and demo applications

Hosting and resource **paradigms**

Closed models are centralized and resource-consuming



Source: numind.ai



- Big players invest billions (Microsoft/OpenAI, AWS/Anthropic)
- CSP selling shovels in the AI Gold rush

Hosting and resource **paradigms**







- **Democratizing** AI Computing
  - Quantization, AI Chips
  - Run models locally, in containers
- Emergence of **smaller models** for edge and mobile
  - Small/Tiny Language Models: Gemini nano, Microsoft Phi-2, Huawei TinyBERT
  - Domain Specific Language Models: BloombergGPT, Harvey (law)
  - Mixture of models: Mixtral 8x7B, OpenMoE → Mixture of licenses?

Key takeaways



- **Hyper-centralization** leads to black boxes and closed solutions
- **Openness**
  - Fosters collaboration and fuels community-driven innovation
  - Enables inclusivity
- Just like open source software beware of **licenses and restrictions**
- **AI's democratization** continually reshapes the landscape

Thank you

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Check the two-part article co-written with Luxin Zhang

Image **credits**

- *Opensource, Internet & GenAI evolution* image generated with DALL-E
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